Data Cleaning

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Our research question is "Does air quality have a statistically significant effect on income inequality within Canadian Metropolitan Areas?"

We've collected our Air Quality Health Index (AQHI) data. We now need to identify the Census Metro Areas (CMAs) associated with each Station ID.

First, let's change our AQHI data to refer to a city rather than a station code!

```
library(tidyverse) #importing libraries, not all are used currently
library(gtools)
library(modelsummary)
library(ggeffects)
library(data.table)
library(sf)
library(jsonlite)
library(data.table)
library(tmap)
library(ggplot2)
library(terra)
library(dplyr)
library(tidyr)
library(splines2)
```

```r
main_dir <- "C:/Users/zaneb/Documents/Vault/UBC/24W/Term 2/ECON 326/FinalProject" #equivalent

regions <- c("atl", "ont", "pnr", "pyr") #the four areas of data collection in AQHI files

d_AQHI <- lapply(setNames(regions, regions), function(region) { #organize by region
  dir_path <- file.path(main_dir, "Data/AQHI", region) #folder of data
  files <- list.files(path = dir_path, pattern = "\\.csv$", full.names = TRUE) #list of files
  rbindlist(lapply(files, fread), use.names = TRUE, fill = TRUE) #reading all of the folders
})

naps <- st_read(file.path(main_dir, "Data/aqhi_station.geojson")) #read the geojson guide to
```

```
Reading layer `aqhi_station' from data source
  `C:\Users\zaneb\Documents\Vault\UBC\24W\Term 2\ECON 326\FinalProject\Data\aqhi_station.geoj
  using driver `GeoJSON'
Simple feature collection with 135 features and 5 fields
Geometry type: POINT
Dimension:     XY
Bounding box:  xmin: -135.0808 ymin: 42.28306 xmax: -52.72528 ymax: 68.3609
Geodetic CRS:  WGS 84
```

```r
valid_cgndb <- unlist(lapply(regions, function(region){
  setdiff(names(d_AQHI[[region]]), c("Date", "Hour (UTC)")) #generate a list of cgndb values
}))
```

We need to adjust that data to correspond with different CMAs.

```r
CMA <- st_read(file.path(main_dir, "Data/Spatial/CMA_shape.shp")) #import CMA
```

```
Reading layer `CMA_shape' from data source
  `C:\Users\zaneb\Documents\Vault\UBC\24W\Term 2\ECON 326\FinalProject\Data\Spatial\CMA_shape
  using driver `ESRI Shapefile'
Simple feature collection with 156 features and 8 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: 3856867 ymin: 687427.7 xmax: 8996148 ymax: 3518467
Projected CRS: NAD83 / Statistics Canada Lambert
```

```
PR <- st_read(file.path(main_dir, "Data/Spatial/PR_shape_simplified.shp")) #import PR
```

```
Reading layer `PR_shape_simplified' from data source
  `C:\Users\zaneb\Documents\Vault\UBC\24W\Term 2\ECON 326\FinalProject\Data\Spatial\PR_shape_
  using driver `ESRI Shapefile'
Simple feature collection with 13 features and 8 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: 3690285 ymin: 687428.3 xmax: 9015165 ymax: 5241994
Projected CRS: NAD83 / Statistics Canada Lambert
```

```
CMA <- CMA |> filter(PRUID != "24") #filter out QC: no data available

naps <- st_transform(naps, 3347) #CRS correction
```

```
naps_CMA <- naps |> filter(lengths(st_intersects(naps, CMA)) > 0) #create a naps dataframe th
naps_CMA <- st_join(naps_CMA, CMA["CMANAME"], join = st_intersects) #add CMA value

naps_df <- as.data.table(naps_CMA)[, .(cgndb, CMANAME, name, longitude = st_coordinates(geome

naps_df <- naps_df[naps_df$cgndb %in% valid_cgndb] #remove irrelevant stations

naps_df[, name := lapply(name, function(x) fromJSON(x)$en)] #adjust names to english

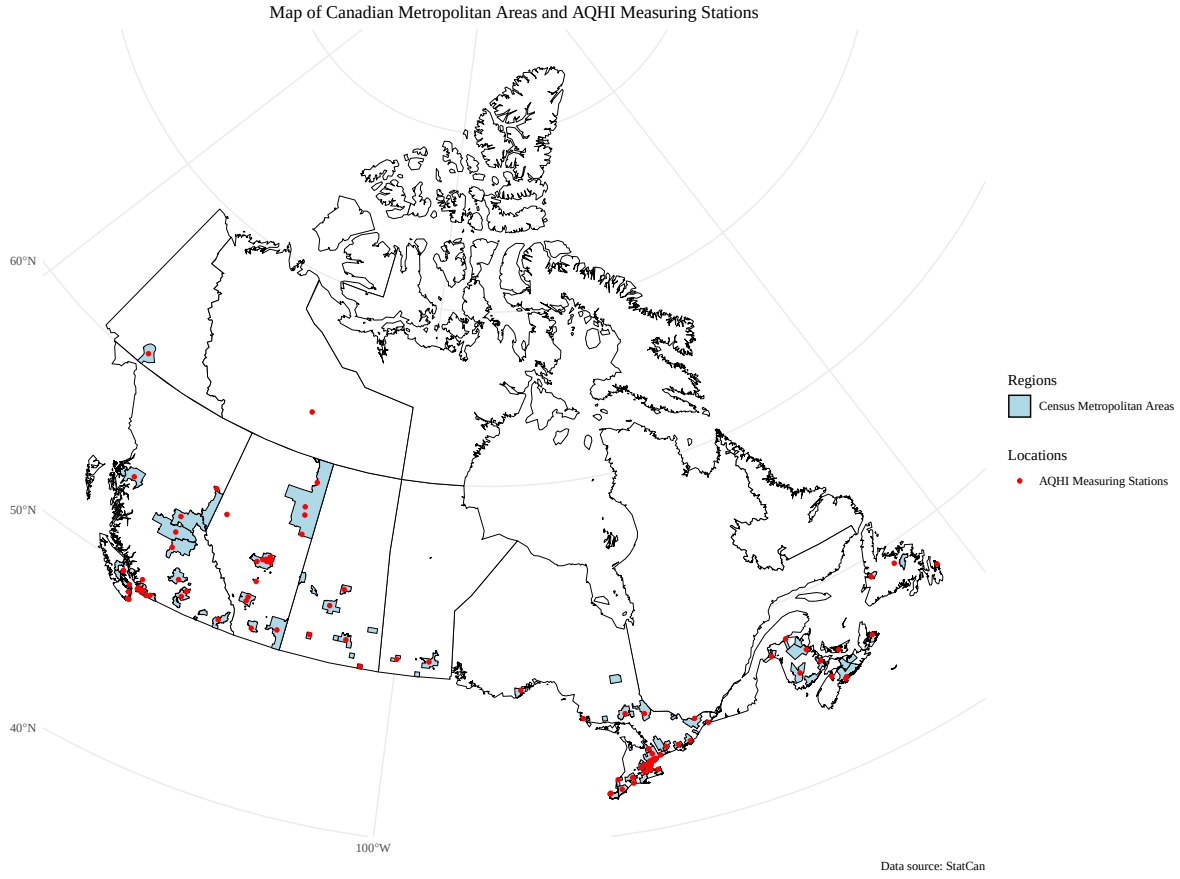
cgndb_key <- setNames(naps_df$name, naps_df$cgndb) #create a key of place names and cgndb val
```

```
placenames <- unlist(lapply(regions, function(region){
  setdiff(names(d_AQHI[[region]]), c("Date", "Hour (UTC)")) #making a list of cgndb now in d
}))
```

Let's map this data.

```
ggplot() +
  geom_sf(data = PR, fill = "white", color = "black") + #PR shapes
  geom_sf(data = CMA, aes(fill = "Census Metropolitan Areas"), color = "black") + #CMA shapes
  geom_sf(data = naps_CMA, aes(color = "AQHI Measuring Stations"), size = 1.2) + #naps points
  scale_fill_manual(name = "Regions",
                    values = c("Census Metropolitan Areas" = "lightblue")) +
  scale_color_manual(name = "Locations",
                     values = c("AQHI Measuring Stations" = "red")) +
```

```
  theme_minimal(base_family = "serif") +
  theme(plot.title = element_text(hjust = 0.5)) + #center title
  labs(title = "Map of Canadian Metropolitan Areas and AQHI Measuring Stations",
       caption = "Data source: StatCan") +
  coord_sf(xlim = c(3690800, 9012300), ylim = c(687400, 5242000)) #set bounds to make sure ma
```



Now let's import our census data.

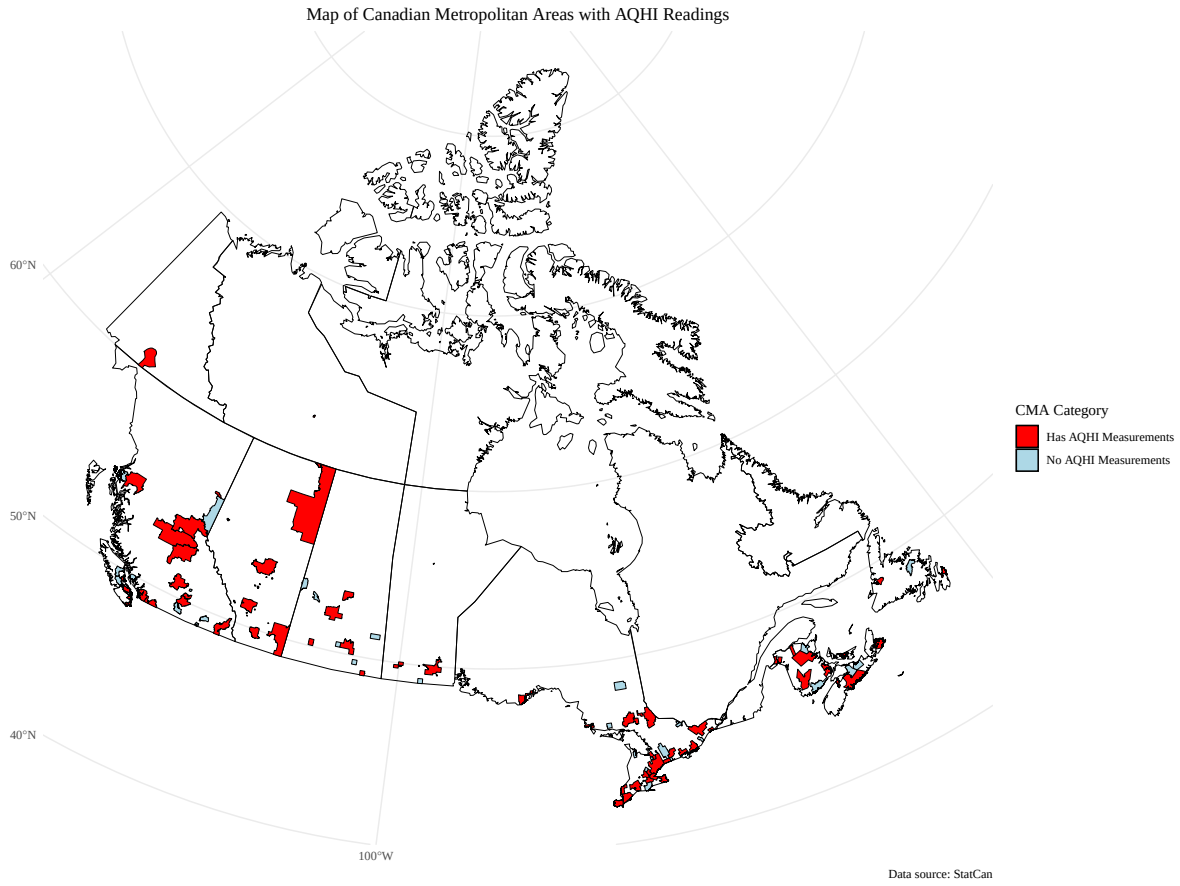
```
d_Census <- fread(file.path(main_dir, "Data/Census/CMA_data_updated.csv")) #import census dat
```

We now need to merge all of our data together into one clear dataframe.

```r
CMA_naps <- CMA |> filter(lengths(st_intersects(CMA, naps)) > 0) #create a CMA dataframe that

CMA_nonaps <- CMA |> filter(lengths(st_intersects(CMA, naps)) == 0) #create a CMA with sites
```

```r
ggplot() +
  geom_sf(data = PR, fill = NA, color = "black") + #PR shapes
  geom_sf(data = CMA_nonaps, aes(fill = "No AQHI Measurements"), color = "black") + #CMA w/o
  geom_sf(data = CMA_naps, aes(fill = "Has AQHI Measurements"), color = "black") + #CMA w/ na
  scale_fill_manual(name = "CMA Category",
                    values = c("No AQHI Measurements" = "lightblue",
                               "Has AQHI Measurements" = "red")) +
  theme_minimal(base_family = "serif") + #theme, font
  theme(plot.title = element_text(hjust = 0.5)) + #center title
  labs(title = "Map of Canadian Metropolitan Areas with AQHI Readings",
       caption = "Data source: StatCan") +
  coord_sf(xlim = c(3690800, 9012300), ylim = c(687400, 5242000)) #scale
```




```
naps_CMA <- st_join(naps_CMA, CMA_naps["CMAUID"], join = st_intersects)#add CMAID values to r

d_CMA <- as.data.table(naps_CMA)[, .(cgndb, CMAUID, CMANAME)] #short df of cdngb values with

aqhi_cgndb <- unlist(lapply(regions, function(region){
  setdiff(names(d_AQHI[[region]]), c("Date", "Hour (UTC)")) #making a list of placenames now
}))
```

```
d_AQHI <- rbindlist(d_AQHI, use.names = TRUE, fill = TRUE, idcol = "region") #break apart reg
```

```
d_AQHI_l <- melt(d_AQHI, id.vars = c("region", "Date", "Hour (UTC)"), variable.name = "Static
```

```r
d <- d_AQHI_l |>
  inner_join(d_CMA, by = c("Station" = "cgndb")) #add CMA data: stations will be labelled by
```

```r
d <- d |>
  group_by(CMAUID, CMANAME, Date) |>
  summarise(mean_AQHI = mean(AQHI, na.rm = TRUE), .groups = "drop") #regroup by average daily
```

```r
d <- d |>
  group_by(CMAUID, CMANAME) |> #group by CMA
  summarise(
    total_days = n(), #total days with data
    high_AQHI_days = sum(mean_AQHI > 3, na.rm = TRUE), #number of days with high AQHI
    percent_high_AQHI = (high_AQHI_days/total_days)*100, #% of days with high AQHI
    average_AQHI = sum(mean_AQHI, na.rm = TRUE)/total_days #average of daily AQHI averages
  )
```

```r
d_Census <- d_Census |>
  rename(CMAUID=`Census Metro Area ID`) #change for merge

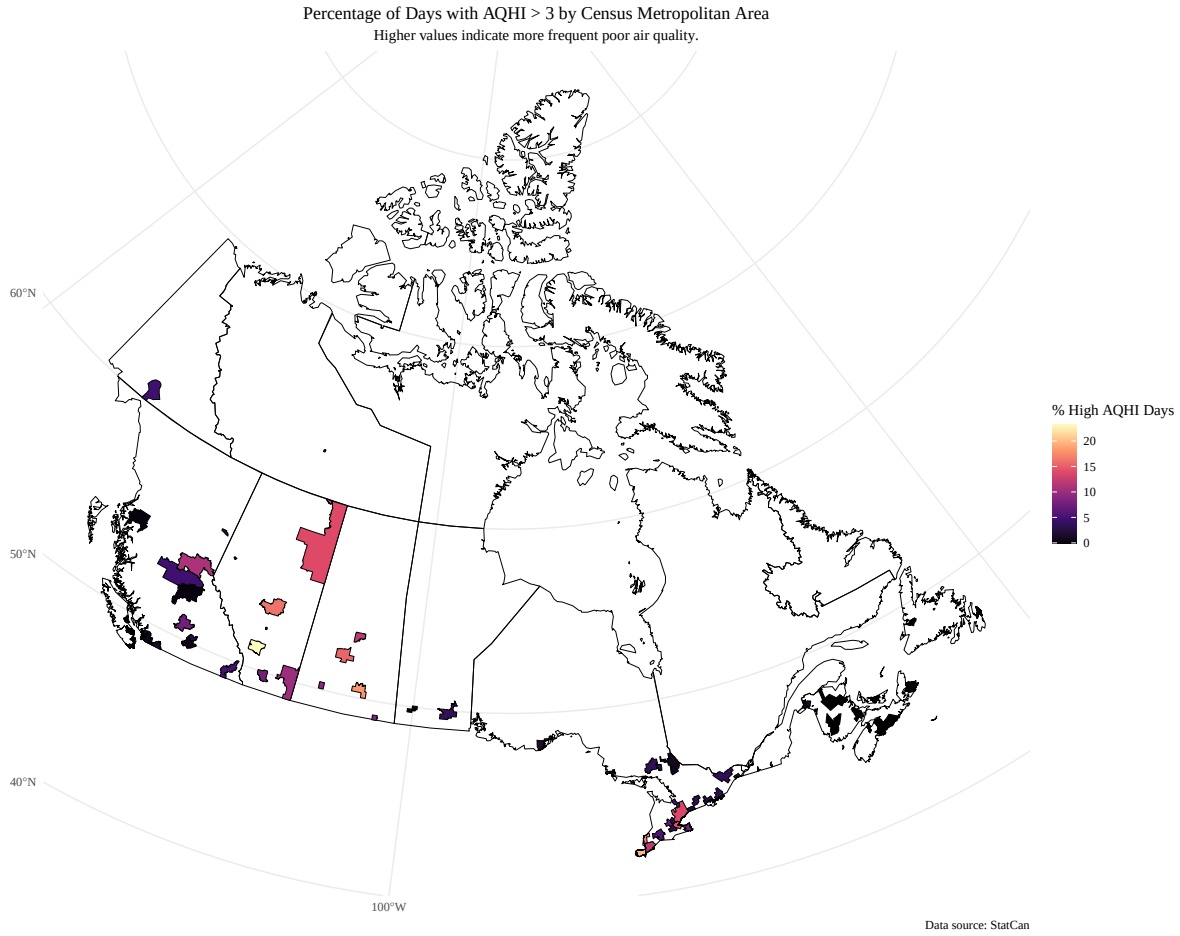
d_Census <- d_Census |>
  mutate(CMAUID = str_pad(as.character(CMAUID), width = 3, side = "left", pad = "0")) #fix fo
```

```r
d <- d |>
  left_join(d_Census, by = "CMAUID") #combine the census and AQHI data!
```

```r
d <- d |>
  left_join(CMA_naps, by = "CMAUID") |> #add the geographic data!
  select(-CMANAME.y) |>
  rename(CMANAME = CMANAME.x) #removing doubled CMANAME column
```

```r
d <- st_as_sf(d) #make d recognized as spatial again
```

```r
ggplot() +
  geom_sf(data = PR, fill = NA, color = "black", size = 0.8) +
  geom_sf(data = d, aes(fill = percent_high_AQHI), color = "black", size = 0.2) +  #color by
  scale_fill_viridis_c(name = "% High AQHI Days", option = "magma") + #color scale
  theme_minimal(base_family = "serif") +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle = element_text(hjust = 0.5)) +
  labs(title = "Percentage of Days with AQHI > 3 by Census Metropolitan Area",
       subtitle = "Higher values indicate more frequent poor air quality.", caption = "Data s
  coord_sf(xlim = c(3690800, 9012300), ylim = c(687400, 5242000))
```



Now we can work with the data! Let's begin by renaming variables.

```
colnames(d)[8] <- "I0" #naming by I matrix as described in our math analysis
colnames(d)[9] <- "I1"
colnames(d)[10] <- "I2"
colnames(d)[11] <- "I3"
print(colnames(d)) #check the change
```

```
[1] "CMAUID"
[2] "CMANAME"
[3] "total_days"
[4] "high_AQHI_days"
[5] "percent_high_AQHI"
[6] "average_AQHI"
```

```
 [7] "Name"
 [8] "I0"
 [9] "I1"
[10] "I2"
[11] "I3"
[12] "% Male Population by Age | Males, 2024"
[13] "% Household Population by Visible Minority Status | Household Population For Visible Mi
[14] "% Household Population 15 Years or Over by Marital Status | Household Population 15 Yea
[15] "% Household Population by Knowledge of Official Language | Household Population For Kno
[16] "% Household Population by Knowledge of Official Language | Household Population For Kno
[17] "% Household Population by Knowledge of Official Language | Household Population For Kno
[18] "% Household Population by Mother Tongue | Household Population For Mother Tongue | Mult
[19] "% Household Population by Total Immigrants and Place of Birth | Total Household Populat
[20] "% Basics | Total Household Population 15 Years Or Over In The Labour Force, 2024"
[21] "% Household Population 15 Years or Over by Labour Force Activity | Employment Rate, 202
[22] "% Household Population by Income (Current Year) | Household Population 15 Years Or Over
[23] "% Household Population 15 Years or Over by Educational Attainment | Household Populatio
[24] "% Household Population 15 Years or Over by Educational Attainment | Household Populatio
[25] "CMAPUID"
[26] "DGUID"
[27] "DGUIDP"
[28] "CMATYPE"
[29] "LANDAREA"
[30] "PRUID"
[31] "geometry"
```

We now need to generate our Gini Coefficient calculations. Let's start by generating the IP matrix.

```
placeholder_matrix <- matrix(NA, nrow = 2, ncol = 5, dimnames = list(c("x", "y"), paste0(1:5)
d$IP <- replicate(nrow(d),placeholder_matrix,simplify = FALSE) #add the matrix to all rows of
```

```
fIP <- function(i){ #iterable
  mat <- d$IP[[i]] #grab empty IP matrix
  I0 <- d$I0[i]
  I1 <- d$I1[i]
  I2 <- d$I2[i]
  I3 <- d$I3[i] #grab I vector values
  xvals <- sort(c(0,0.5,1-(I1/100),1-(I1*I2/10000),1-(I1*I2*I3/1000000))) #calculate IP matri
  yvals <- sort(c(0,I0/300000,100000/300000,200000/300000,1)) #calculate IP matrix y values
  mat <- matrix(c(xvals, yvals), nrow = 2, byrow = TRUE) #assign values to matrix
```

```
  return(mat) #assign matrix to d$IP
}

d$IP <- lapply(seq_len(nrow(d)), fIP) #apply the function above to all rows in d

fix_IP <- function(mat){
  mat[2, ] <- pmin(mat[2, ],mat[1, ])  # force IPy to be less than or equal to IPx for income
  return(mat)
}
d$IP <- lapply(d$IP, fix_IP)

d$P <- lapply(d$IP, function(mat){ #list apply to create P function for all rows in d
  splinefun(mat[1,],mat[2,],method = "hyman") #cubic spline function, using "hyman" method fo
})

d$E <- lapply(d$IP, function(mat){ #list apply to create E function for all rows in d
  slope <- mat[2,5]/mat[1,5]  # IPy[5]/IPx[5]
  return(function(x) slope*x) #return function
})

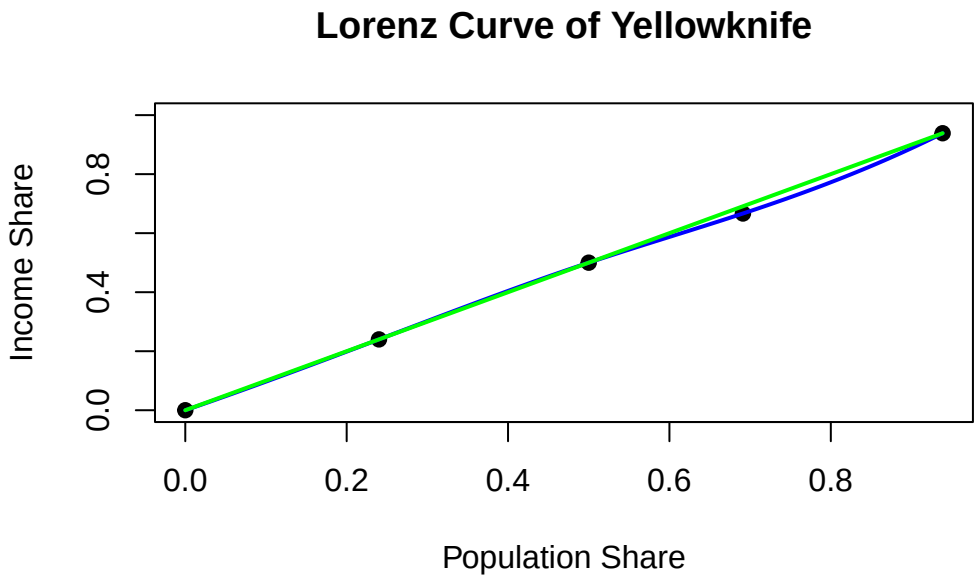
d$EP <- lapply(seq_len(nrow(d)), function(i){ #list apply to create EP function for all rows
  E <- d$E[[i]]
  P <- d$P[[i]]
  return(function(x) E(x) - P(x)) #return function
})

d$Gini <- sapply(seq_len(nrow(d)), function(i){ #numerical values
  areaA <- integrate(d$EP[[i]],d$IP[[i]][1,1],d$IP[[i]][1,5])
  areaB <- integrate(d$E[[i]],d$IP[[i]][1,1],d$IP[[i]][1,5])
  return(areaA$value/areaB$value) #will assign this value to all rows of d$Gini
})

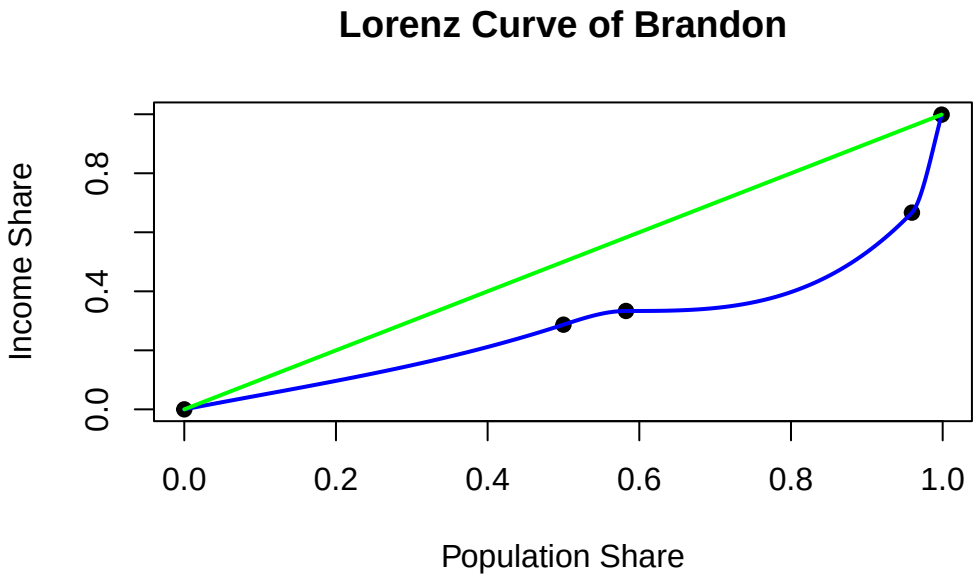
Lorenz <- function(i){ #function to plot Lorenz curves: some Gini coefficients were returning
  IPx <- d$IP[[i]][1,]
  IPy <- d$IP[[i]][2,]
  P <- d$P[[i]]
  E <- function(x) (IPy[5] / IPx[5]) * x
  x <- seq(IPx[1], IPx[5], length.out = 1000) #linspace of x
  y  <- P(x) #approximation spline
  Ey <- E(x) #equality line
```

```
  plot(IPx, IPy, pch = 19, main = paste("Lorenz Curve of", d$CMANAME[i]),
       xlab = "Population Share", ylab = "Income Share", ylim = c(0, 1))
  lines(x, y, col = "blue", lwd = 2)
  lines(x, Ey, col = "green", lwd = 2) #plotting
}
```

```
Lorenz(66) #example of a high-income CMA where the Gini coefficient is hard to estimate with
```




```
Lorenz(35) #example of a high-income CMA where the Gini coefficient is easier to estimate wi
```


Lorenz Curve of Brandon
Income Share
0.0
0.4
0.8
0.0
0.2
0.4
0.6
0.8
1.0
Population Share